

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Roadmap

01 Background

Multi-page, visually rich document understanding: what the documents are, what the task is, and how the field got here.

02 Survey

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

EMNLP 2026, main conference

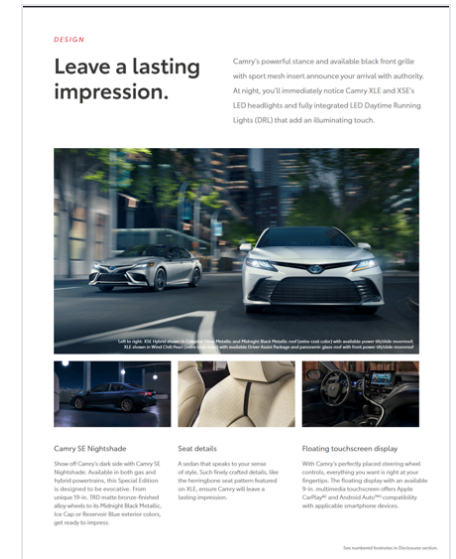
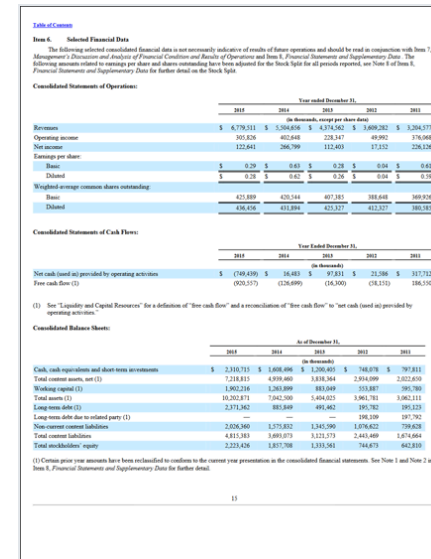
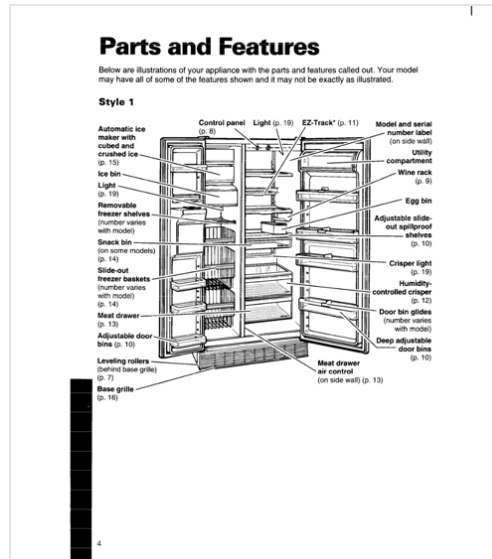
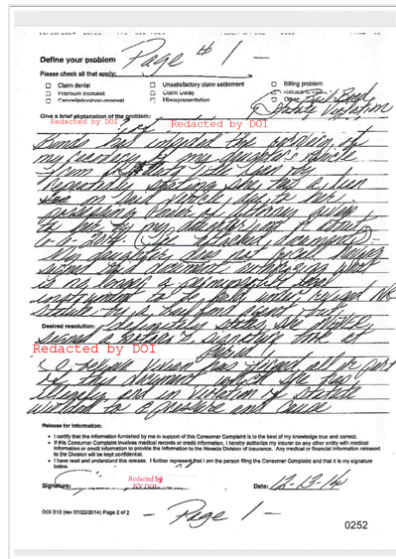
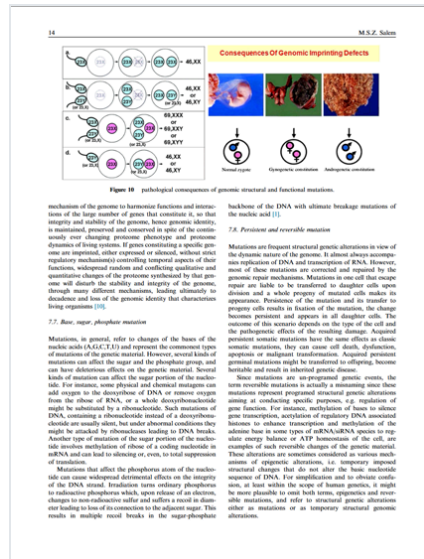
03 Empirical attribution study

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

Submitted to EACL, under review

Multi-Page Visually Rich Documents

Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.



Research paper
born-digital

Signed form
scanned, handwritten

Technical manual
layout-dependent

Financial report
dense tables

Product brochure
photo-led

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

One broad way to define it: get a system to **answer a natural-language question over a document**, accurately. The systems range from rule-based lookup, through trained vision-language models, to pipelines built around pretrained general-domain ones.

- 1

"What year was this report published?"

FACT LOOKUP
- 2

"According to the table, which region had the highest revenue in 2023?"

READ A TABLE
- 3

"How much did the leading region's revenue grow between the 2022 and 2023 results tables?"

TWO TABLES, THEN ARITHMETIC
- 4

"Does the growth described in the executive summary agree with the figures in the results table?"

PROSE AGAINST NUMBERS
- 5

"Across the report and its appendix, what evidence supports the conclusion that the new method improved performance, and what limitation qualifies it?"

GATHER, THEN QUALIFY

Illustrative questions, not quoted benchmark items

A Brief History

A useful starting point is **MP-DocVQA** in 2023: a benchmark of documents up to twenty pages, with questions answerable from a single page, plus the Hi-VT5 baseline. It is the pivotal moment at which long-document understanding became its own problem.

BENCHMARKS

MP-DocVQA, 2023

Up to twenty pages. One page carries the answer, so the task is finding it.

MMLongBench-Doc, LongDocURL

- Documents into the hundreds of pages, with questions that combine evidence from several places at once.

MODELS

Purpose-built, trained end to end

Hi-VT5 and its successors: a document encoder trained for this setting specifically.

General-domain VLMs as the backbone

- Increasingly sophisticated training and training-free machinery built around a pretrained model that already reads text and page images well.

A lot happened in between, and quickly. Organising that is what the survey does.

Document Questions in Consumer Assistants

Everyone here has used these. What changed is not the question being asked, but **how much of the document the model has to hold at once**.

A YEAR OR TWO AGO

The whole document, in context

Parsed text plus an image of every page, all placed in the model's context window. Fine for short documents, and it hits the ceiling fast.

SCREENSHOT
"cannot upload document,
out of context" error

MORE RECENTLY

On disk, read by tool calls

The upload lands in the model's working environment. It reads what it needs, when it needs it, rather than ingesting everything up front.

SCREENSHOT
assistant reading an uploaded
file through a tool call

AND NOW

Projects, searched by retrieval

Upload many documents at once. Beyond what fits in context, the content is indexed and retrieved against the question instead.

SCREENSHOT
a project with several
uploaded documents

Real-World Constraints

A hospital, a mining company, or a law firm wants the same thing: **classification, summarisation, or structured extraction over tens of thousands of documents**. The consumer path is often not available to them.

WHY NOT JUST CALL AN API

- **The documents cannot leave.** Medical records, legal and financial filings, and resources data carry privacy and data-residency requirements that rule out a third party.
- **The volume makes it uneconomic.** At tens of thousands of documents, per-call pricing stops being worth the benefit even where sending the data is permitted.
- **The document type is unusual.** A niche form or report may need its own training or fine-tuning rather than whatever a general model already knows.

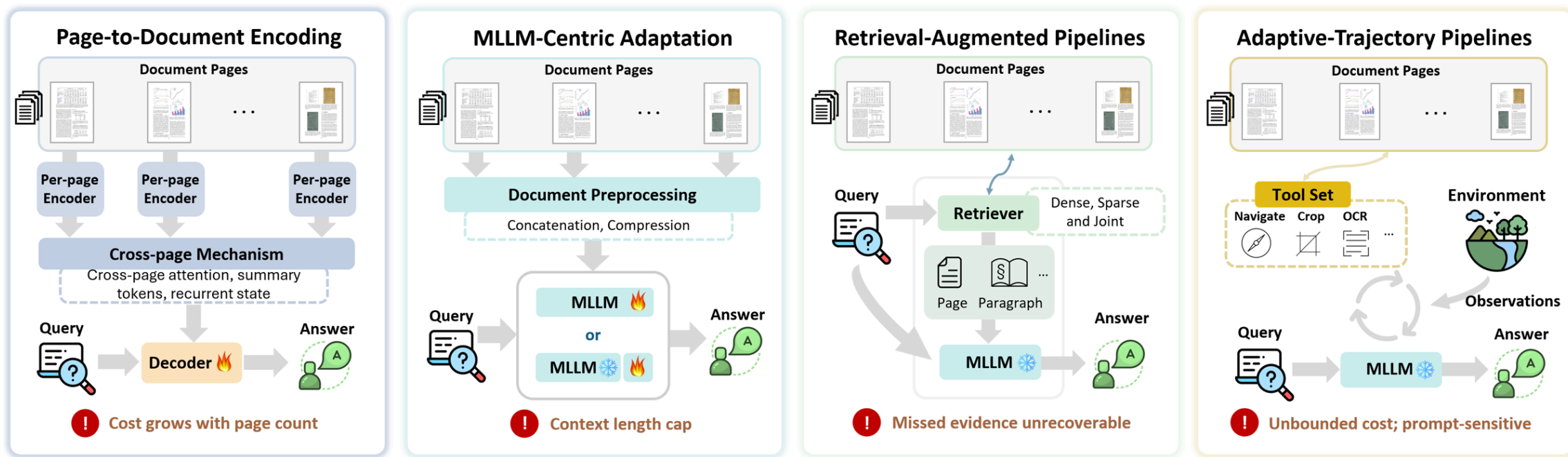
WHAT THAT FORCES

- **Open-weight models, run locally.** A pretrained vision-language model on consumer hardware, not a datacentre.
- **A fixed compute budget.** Which pages are read, at what resolution, and how many times, all have to fit inside it.
- **Efficiency as a research problem.** Cost and latency are open questions in their own right, not implementation details.

This is where the research sits: **techniques and architectures for document understanding under a fixed compute budget**. The same techniques then flow back into the consumer products, which is why those have moved so quickly.

Multi-Page Architectural Overview

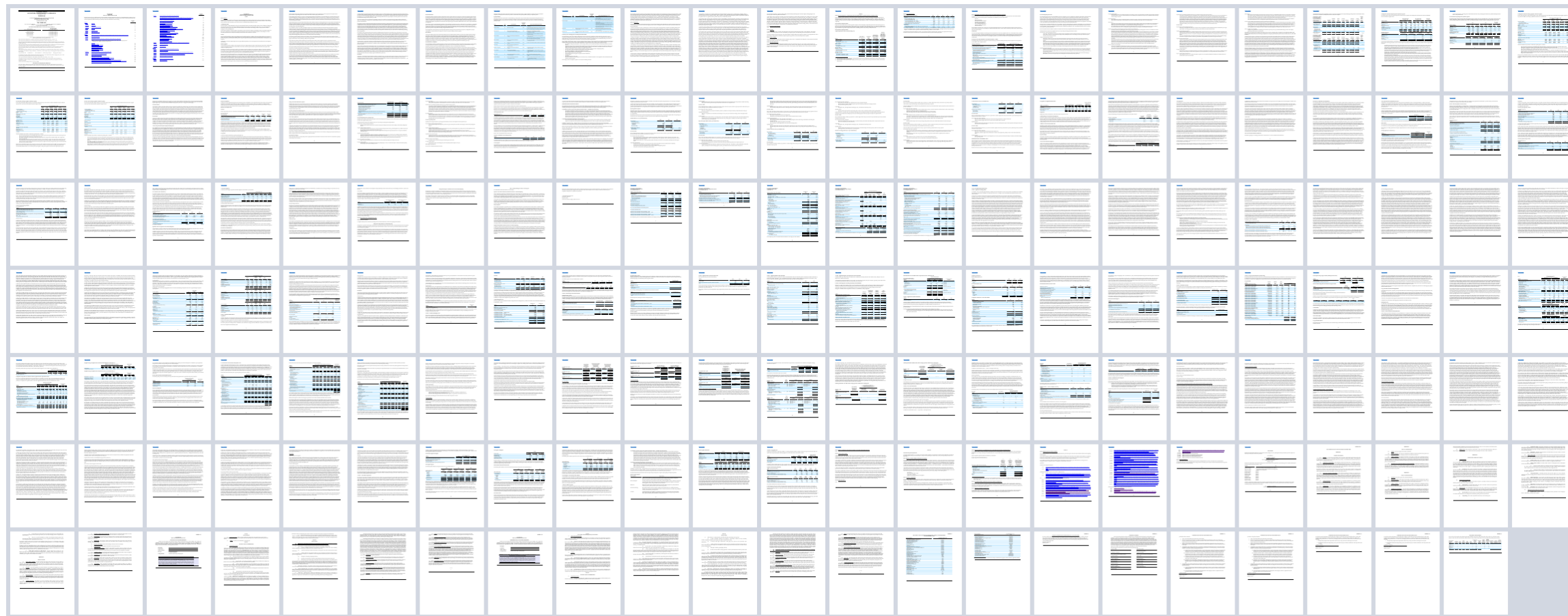
The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM's context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

Text Modality

VOLUME one annual report: 160 pages, about 89,000 words



Text Modality

CONTINUITY a sentence cut by the page

Despite the limitations of our work, we find our results to be highly encouraging. We show that substantial weak-to-strong generalization is not only possible, but actually a widespread phenomenon. We also show that with very simple methods, we can drastically improve the ability of weak supervisors to elicit knowledge from strong models. With much more progress in this direction, we could get to the point where we can use weak supervisors to reliably elicit knowledge from much stronger

3

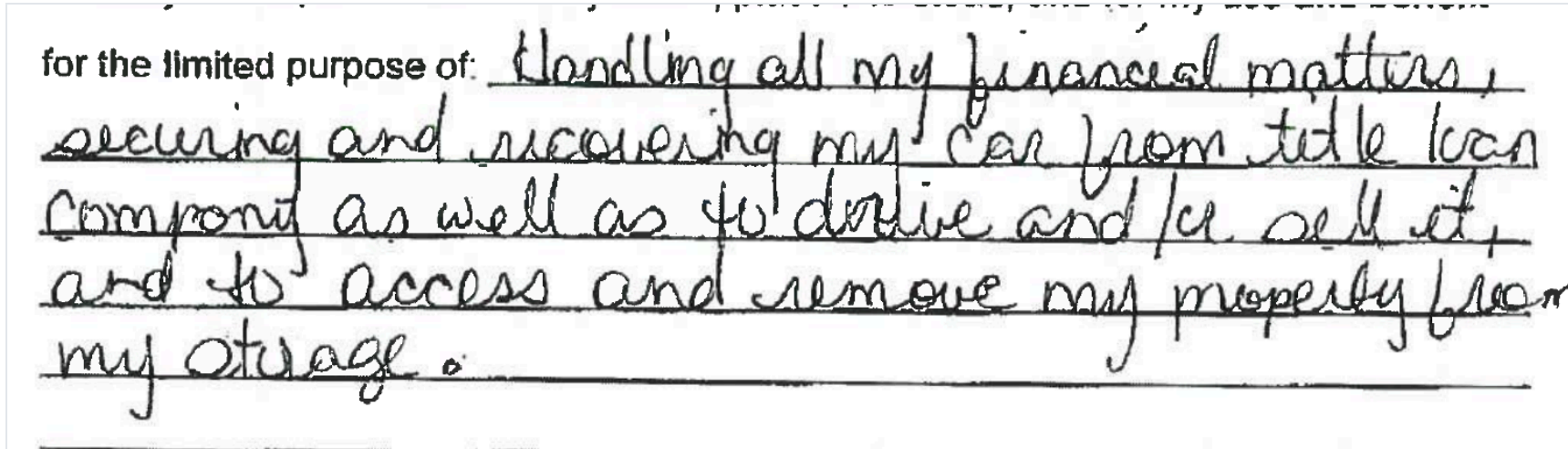
----- PAGE 3 ENDS, PAGE 4 STARTS -----

models, at least for some key tasks that we care about. This may allow us to develop superhuman reward models or safety classifiers, which we could in turn use to align superhuman models.

Aligning superhuman models is essential for making them safe; there is increasing recognition that failing to align such powerful models has the potential to be catastrophic, making this one of the most important unsolved technical problems in the world (CAIS). We think it is now more tractable than ever to make rapid iterative empirical progress toward solving this problem.

Text Modality

NOISE what OCR reads back



The scan · a handwritten power of attorney

for the limited purpose of: _ Liane in OA Wy J A anual mallins
QLEUIINA OO DALCOLON IDO Wi 'On LfLiorv AS iC
mond OQ oN Oo Yo! OWE end |e ud M_ Lk
AY RACDAS GIA AMM Ore noah were rly f-As0-,
ea} OL OG0

The OCR output · this is also what a text retriever would index

Text Modality

HETEROGENEITY same report, two pages

Table of Contents
Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.
/s/ PricewaterhouseCoopers LLP Minneapolis, Minnesota February 7, 2019
We have served as the Company's auditor since 1975.
55

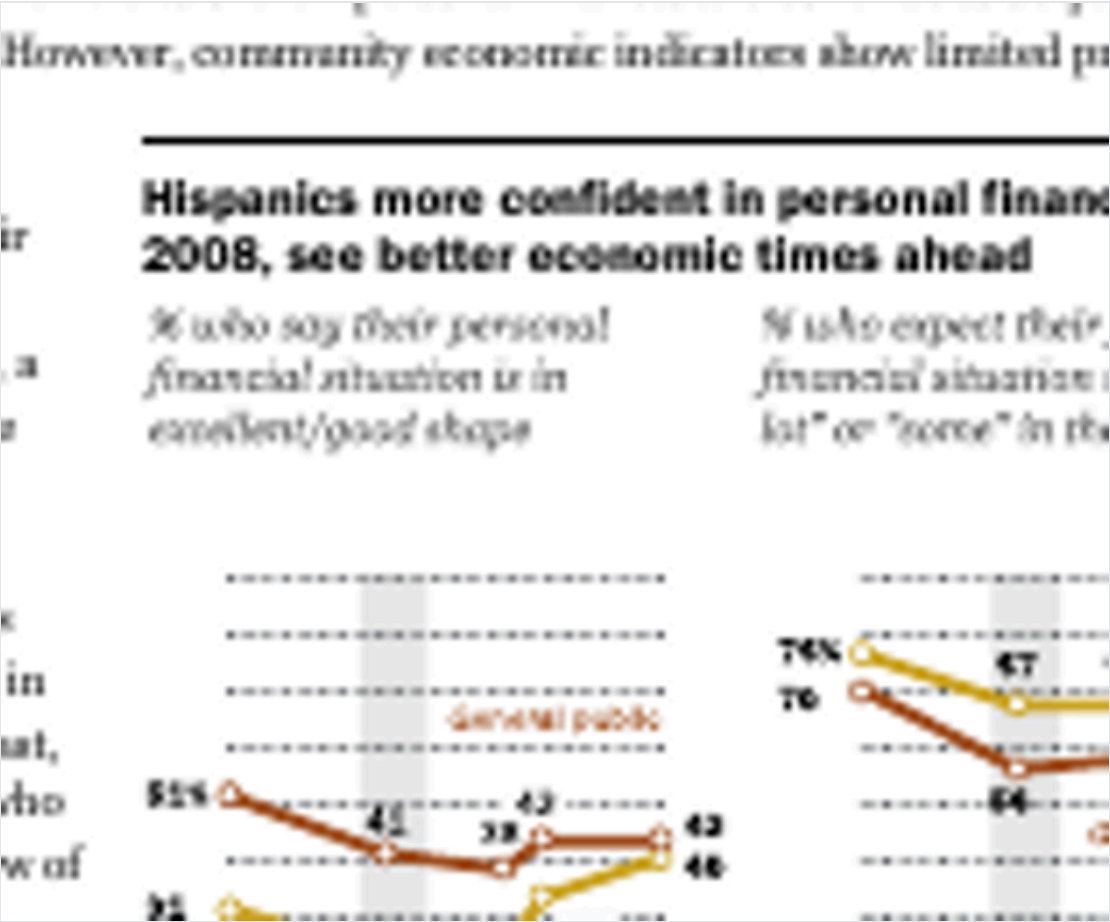
Page 55 • 75 words

Table of Contents
certain other instruments, including trade and other receivables, loans, held-to-maturity debt securities, net investments in leases and off-balance-sheet credit exposures. With respect to available-for-sale (AFS) debt securities, the ASU amends the current other-than-temporary impairment model. For such securities with unrealized losses, entities will still consider if a portion of any impairment is related only to credit losses and therefore recognized as a reduction in income. However, rather than also reflecting that credit loss amount as a permanent reduction in cost (amortized cost) basis of that AFS debt security, the ASU requires that credit losses be reflected as an allowance. As a result, under certain circumstances, a recovery in value could result in previous allowances, or portions thereof, reversing back into income. For 3M, this ASU is effective January 1, 2020, with early adoption permitted. Entities are required to apply the standard's provisions as a cumulative-effect adjustment to retained earnings as of the beginning of the first reporting period in which the guidance is adopted. The Company is currently assessing this ASU's impact on 3M's consolidated result of operations and financial condition.
In October 2016, the FASB issued ASU No. 2016-16, <i>Extra-Entity Transfers of Assets Other Than Inventory</i> , which modifies existing guidance and is intended to reduce diversity in practice with respect to the accounting for the income tax consequences of intra-entity transfers of assets. The ASU indicates that the current exception to income tax accounting that requires companies to defer the income tax effects of certain intercompany transactions would apply only to intercompany inventory transactions. That is, the exception would no longer apply to intercompany sales and transfers of other assets (e.g., intangible assets). Under the existing exception, income tax expense associated with intra-entity profits in an intercompany sale or transfer of assets is eliminated from earnings. Instead, that cost is deferred and recorded on the balance sheet (e.g., as a prepaid asset) until the assets leave the consolidated group. Similarly, the entity is prohibited from recognizing deferred tax assets for the increases in tax bases due to the intercompany sale or transfer. For 3M, the ASU was effective January 1, 2018 using the modified retrospective method of adoption. Prior periods have not been restated and the January 1, 2018 balance of retained earnings was decreased by less than \$2 million.
In January 2017, the FASB issued ASU No. 2017-01, <i>Clarifying the Definition of a Business</i> , which narrows the existing definition of a business and provides a framework for evaluating whether a transaction should be accounted for as an acquisition (or disposal) of assets or a business. The ASU requires an entity to evaluate if substantially all of the fair value of the gross assets acquired is concentrated in a single identifiable asset or a group of similar identifiable assets; if so, the set of transferred assets and activities (collectively, the set) is not a business. To be considered a business, the set would need to include an input and a substantive process that together significantly contribute to the ability to create outputs. The standard also narrows the definition of outputs. The definition of a business affects areas of accounting such as acquisitions, disposals and goodwill. Under the new guidance, fewer acquired sets are expected to be considered businesses. For 3M, this ASU was effective January 1, 2018 on a prospective basis and the Company will apply this guidance to applicable transactions after the adoption date.
In February 2017, the FASB issued ASU No. 2017-05, <i>Clarifying the Scope of Asset Derecognition Guidance and Accounting for Partial Sales of Nonfinancial Assets</i> . This ASU addresses scope-related questions that arose after the FASB issued its revenue guidance in ASU No. 2014-09, <i>Revenue from Contracts with Customers</i> . The new standard clarifies the accounting for derecognition of nonfinancial assets and defines what is considered an in substance nonfinancial asset. Nonfinancial assets largely relate to items such as real estate, ships and intellectual property that do not constitute a business. The new ASU impacts entities derecognizing (e.g., selling) nonfinancial assets (or in substance nonfinancial assets), including partial interests therein, when the purchaser is not a customer. Under the new guidance, the seller would apply certain recognition and measurement principles of ASU No. 2014-09, <i>Revenue from Contracts with Customers</i> , even though the purchaser is not a customer. For 3M, this new standard was effective coincident with the Company's January 1, 2018 adoption of ASU No. 2014-09.
In March 2017, the FASB issued ASU No. 2017-07, <i>Improving the Presentation of Net Periodic Pension Cost and Net Periodic Postretirement Benefit Cost</i> . This ASU changes how employers that sponsor defined benefit pension and/or other postretirement benefit plans present the net periodic benefit cost in the income statement. Under the new standard, only the service cost component of net periodic benefit cost would be included in operating expenses and only the service cost component would be eligible for capitalization into assets such as inventory. All other net periodic benefit cost components (such as interest, expected return on plan assets, prior service cost amortization and actuarial gain/loss amortization) would be reported outside of operating income. 3M adopted this ASU is effective January 1, 2018 on a retrospective basis, however, guidance limiting the capitalization to only the service cost component is applied on prospective basis. The Company previously filed a Current Report on Form 8-K dated May 8, 2018 (which updated 3M's 2017 Annual Report on Form 10-K) that provided prior period information reflecting the retrospective adoption of this ASU. The adoption had no impact on previously reported income before income taxes and net income attributable to 3M. However, non-service cost components of net periodic benefit costs in prior periods were reclassified from operating expenses
68

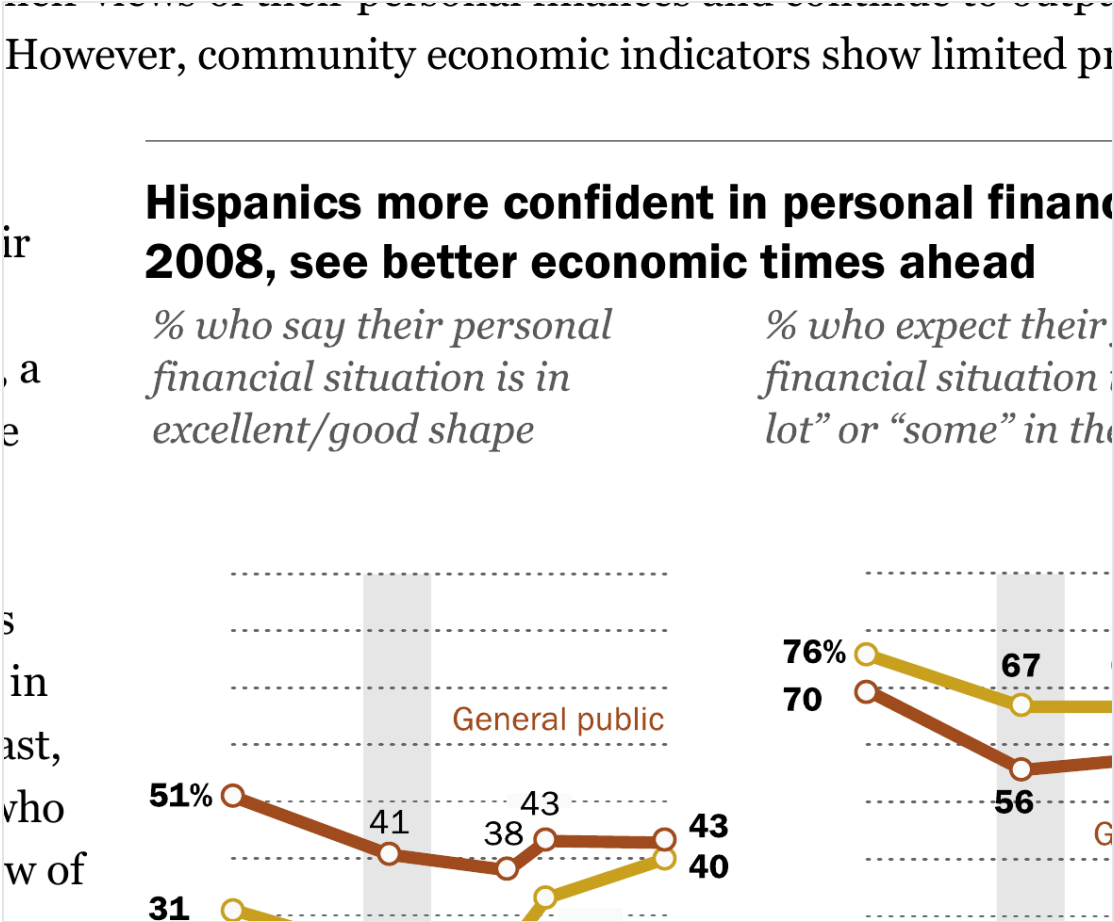
Page 68 • 946 words

Visual Modality

RESOLUTION the same chart at two input resolutions



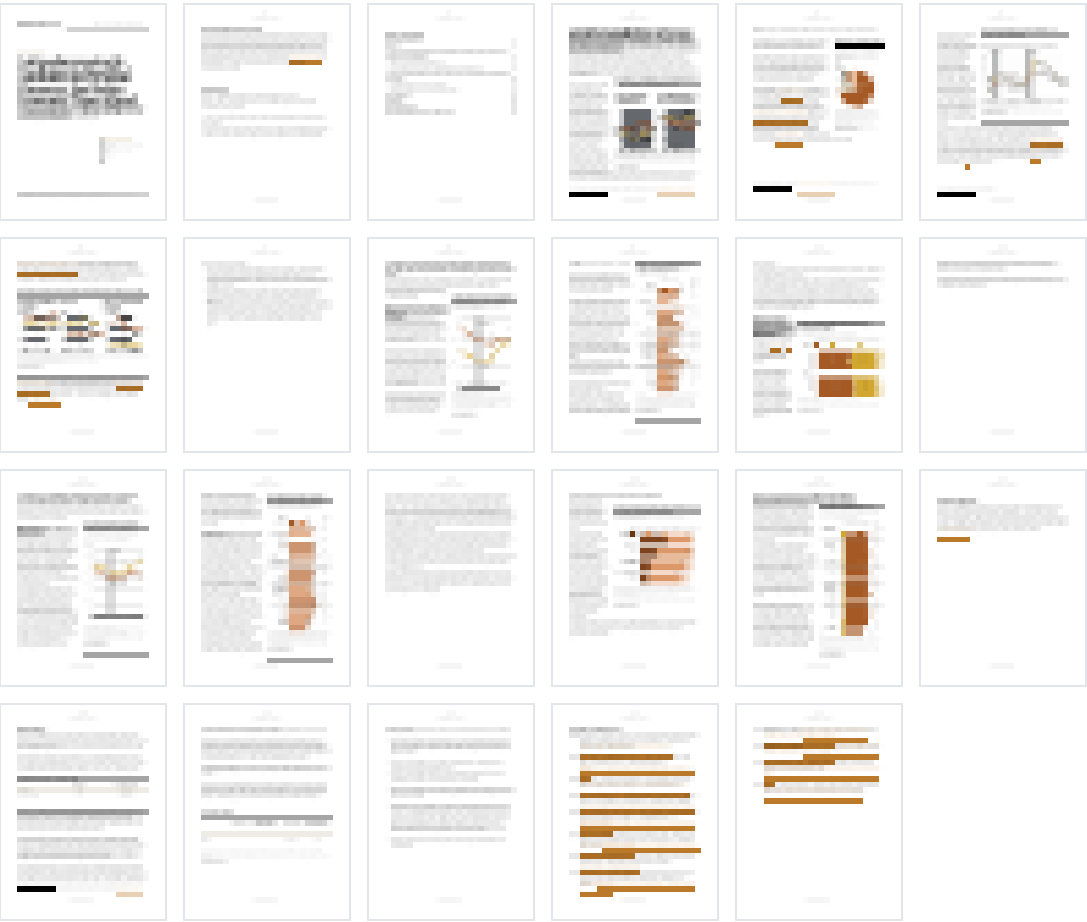
Page at 340 × 440 px · labels gone



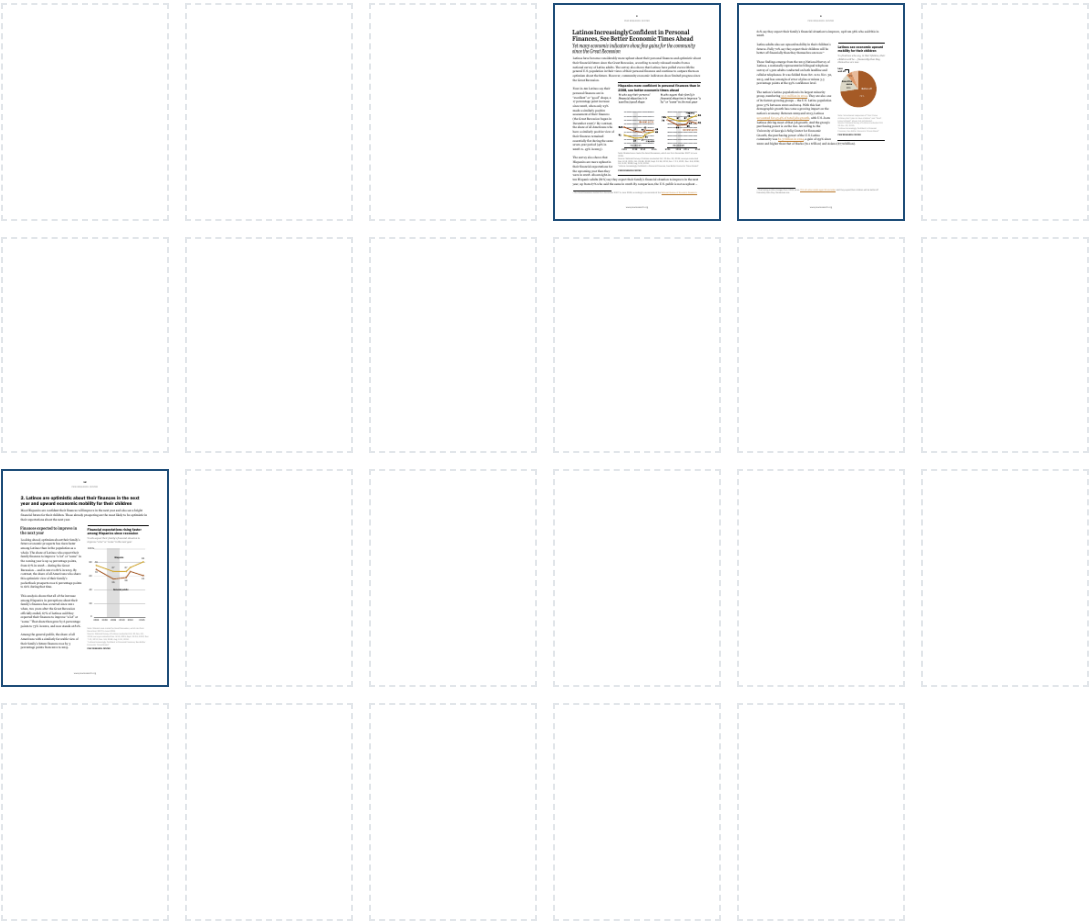
Page at 1275 × 1650 px · 14 times the pixels

Visual Modality

RESOLUTION one 23-page report, the same token budget



Token reduction · every page available, at lower fidelity



Retrieved pages only · high fidelity, capped by the selector's recall

Visual Modality

MULTI-SIGNAL COMPETITION one per-page budget

3

PEW RESEARCH CENTER

Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

TEXT

Latinos say their personal finances are in “excellent” or “good” shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).¹ By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

CHART

confident in personal finances than in 2008, see better economic times ahead

% who say their personal financial situation is in excellent/good shape

Year	General public	Hispanic
2004	41	23
2008	41	24
2011	43	33
2015	43	40

% who expect their family's financial situation to improve “a lot” or “some” in the next year

Year	General public	Hispanic
2004	70	76
2008	67	67
2011	67	73
2015	61	81

Note: Shaded area marks the Great Recession, which ran from December 2007 to June 2009.

Source: National Survey of Latinos conducted Oct. 21-Nov. 30, 2015; surveys conducted Dec. 8-13, 2015; Oct. 24-28, 2012; Sept. 12-16, 2012; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 9-12, 2008; Aug. 5-10, 2004.

“Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead”

PEW RESEARCH CENTER

TEXT

Latinos are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (81%) say they expect their family's financial situation to improve in 2016, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat

TEXT

¹ The Great Recession lasted from December 2007 to June 2009, according to economists at the [National Bureau of Economic Research](#).

www.pewresearch.org

Text-bearing pixels • body prose

Four-in-ten Latinos say their personal finances are in “excellent” or “good” shape, a 17 percentage point increase

Non-textual evidence • a chart

Year	General public	Hispanic
2004	41	23
2008	41	24
2011	43	33
2015	43	40

Year	General public	Hispanic
2004	70	76
2008	67	67
2011	67	73
2015	61	81

Text-bearing pixels again • the source note, in smaller type

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“Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead”

PEW RESEARCH CENTER

Pages from MMLongBench-Doc

15 / 40 • Survey

Cross-Page Structure Modality

HIERARCHY sections and headings organise the whole document

Table of Contents

About Pew Research Center1

Overview3

1. Latinos are increasingly positive about current finances, but many say family income is not keeping pace with cost of living8

Views of current personal finances rebound8

For many Latinos, family income is not keeping pace with the cost of living10

2. Latinos are optimistic about their finances in the next year and upward economic mobility for their children12

Finances expected to improve in the next year12

Latinos see upward economic mobility for their children16

Acknowledgments17

Methodology18

Appendix A: References21

Appendix B: 2015 National Survey of Latinos Topline23

Survey report · two chapters, each with its own subsections

Cross-Page Structure Modality

CROSS-PAGE REFERENCES a pointer to a referent on another page

Page 4 · the sentence

prehension by both the models and humans, since WikiData utilizes QIDs (e.g., Q220) to represent unique entities. We construct a prompt (Table 13 in appendix D) which includes 1) instruction to the models to generate attributed answers. 2) retrieved knowledge graph, and 3) the question. We

p.4 → p.19 fifteen pages later, in an appendix

Page 19 · the caption it means

Table 13: Full instruction and demonstration for answer generation with citaion. We use brown color for the knowledge pool, and blue color for the expected outcome provided by the demonstration.

Page 19

Instruction: You answer the question based on your knowledge, with the given information for annotation, following the given format. Use [NA] for claims that need annotation but is unprovided.

Question: Considering the information:
{name: Orazio Gentileschi, place of death: London, child: Artemisia Gentileschi, notable works: Diana the Huntress, given name: Orazio, surname: Gentileschi, languages spoken: Italian, movement: mannerism, work location: Rome, ethnic group: Italians, date of birth: 1563-07-19, date of death: 1639-02-07, qid: Q367360}
{name: Artemisia Gentileschi, place of birth: Rome, place of death: Naples, occupation: painter, member of: Accademia delle Arti del Disegno, father: Orazio Gentileschi, notable works: Judith Slaying Holofernes, movement: Caravaggisti, ethnic group: Italians, work location: Florence, depicted by: Artemisia, surname: Gentileschi, languages spoken: Italian, student of: Orazio Gentileschi, given name: Artemisia, mother: Prudenzia di Ottaviano Montoni, date of death: 1654-01-01, qid: Q212657}
How did Orazio Gentileschi's influence on Artemisia's life and career shape her development as a prominent Baroque painter, despite facing significant challenges as a female artist in a male-dominated field?

Answer: Artemisia Gentileschi was an Italian painter born on July 8, 1596 [NA] in Rome [Q212657, ethnic group: Italians, occupation: painter, place of birth: Rome]. She was a member of the Accademia delle Arti del Disegno and is best known for her work Judith Slaying Holofernes [Q212657, member of: Accademia delle Arti del Disegno, notable works: Judith Slaying Holofernes]. She was the eldest child of the Tuscan painter Orazio Gentileschi [Q212657, father: Orazio Gentileschi]. Orazio Gentileschi was an Italian painter [NA] born in 1563 and died in 1639 [Q367360, ethnic group: Italians, date of birth: 1563-07-19, date of death: 1639-02-07]. He was born in Pisa and died in London [Q367360, place of death: London]. Orazio's influence on Artemisia's development as a prominent Baroque [NA] painter can be seen in her highly naturalistic portrayal of figures, dramatic scenes and the use of chiaroscuro technique [NA]. He also provided her with the opportunity to study with him and learn from his experience and expertise. She became an important second-generation proponent of Caravaggio's dramatic realism [Q212657, movement: Caravaggisti].

Table 13: Full instruction and demonstration for answer generation with citaion. We use brown color for the knowledge pool, and blue color for the expected outcome provided by the demonstration.

Cross-Page Structure Modality

SPANNING ELEMENTS one table, two pages

topics, texts, and issues, building on others' ideas and expressing their own clearly and persuasively. 5. <u>Speaking and Listening 11/12.2</u>: Integrate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, orally) in order to make informed decisions and solve problems, evaluating the credibility and accuracy of each source and noting any discrepancies among the data. 6. <u>Reading for Informational Text 11/12.3</u>: Analyze a complex set of ideas or sequence of events and explain how specific individuals, ideas, or events interact and develop over the course of the text. 7. <u>Reading for Informational Text 11/12.4</u>: Determine the meaning of words and phrases as they are used in a text, including figurative, connotative, and technical meanings, analyze how an author uses and refines the meaning of a key term or terms over the course of a text. 8. <u>Reading for Informational Text 11/12.7</u>: Integrate and evaluate multiple sources of information presented in different media or formats (e.g., visually, quantitatively) as well as in words in order to address a question or solve a problem. SUMMARY OF ALL KEY ASSIGNMENTS <table><tr><td> QUARTER 1: UNIT 1 Predicting Your Success: TED TALK, GRIT: The Power of Passion and Perseverance; submit a written response on to Google Classroom. Unit 1 & 2 Team Activity Form five to seven teams. Each team is responsible for one of the sectors of the external business environment discussed in the chapter (economic, political/legal, demographic, social, competitive, global, and technological). (Interpersonal, Resources, Information) <u>Quiz #1: Twenty core concepts from Unit 1 & 2</u> Critical Thinking Case Study-- UBER Hits a Bumpy Road.: Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your findings. -(Resources, Interpersonal, Information, Technology) Critical Thinking Case Study-- We Want Our MTV (International) <u>Quiz #2: Twenty Core Concepts on Unit 3 & 4</u> ...Continued Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your Findings -(Resources, Interpersonal, Information, Technology) Preparing for Tomorrow's Workplace Skills: - Do you have what it takes to be a successful franchisee? Working the Net: - Select three franchises that interest you. </td><td> QUARTER 2: Preparing Working the Net: Business Plan - Visit Sample Business Plans at http://www.bplans.com to review examples of all types of business plans. - Select an idea for a company in a field that interests you and, using information from the site, prepare an outline for its business plan. <u>Quiz #3- Comprehension of Twenty Core Concepts from Unit 5 & 6</u> Preparing for Tomorrow's Workplace Skills: Self Reflection (1 paragraph per question submitted into Google Classroom) Unit 7 Ethics Activity-Ethics Report; Training IT Replacement Working the Net Using a search engine, look for the term "company organizational charts," and find at least three examples of organizational charts for corporations, not-for-profits, or government agencies. Unit 8 Key Assignments: <u>Quiz #4-Comprehension of Twenty Core Concepts from Unit 7 & 8</u> <u>Critical Thinking Case Study- Discrimination in the Workplace Continues--submit on Google Docs</u> Semester Final Assessments: Part 1: Ethics Self-Assessment Part 2: Comprehensive Final on Key Concepts from Unit 1-8 - Students will be in 8 teams of 4 assigned to 1 Unit each. - They will choose the key concepts from each unit to teach and add to the final. (A form of Reciprocal Teaching) </td></tr></table>		QUARTER 1: UNIT 1 Predicting Your Success: TED TALK, GRIT: The Power of Passion and Perseverance; submit a written response on to Google Classroom. Unit 1 & 2 Team Activity Form five to seven teams. Each team is responsible for one of the sectors of the external business environment discussed in the chapter (economic, political/legal, demographic, social, competitive, global, and technological). 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Report your Findings -(Resources, Interpersonal, Information, Technology) Preparing for Tomorrow's Workplace Skills: - Do you have what it takes to be a successful franchisee? Working the Net: - Select three franchises that interest you.	QUARTER 2: Preparing Working the Net: Business Plan - Visit Sample Business Plans at http://www.bplans.com to review examples of all types of business plans. - Select an idea for a company in a field that interests you and, using information from the site, prepare an outline for its business plan. <u>Quiz #3- Comprehension of Twenty Core Concepts from Unit 5 & 6</u> Preparing for Tomorrow's Workplace Skills: Self Reflection (1 paragraph per question submitted into Google Classroom) Unit 7 Ethics Activity-Ethics Report; Training IT Replacement Working the Net Using a search engine, look for the term "company organizational charts," and find at least three examples of organizational charts for corporations, not-for-profits, or government agencies. 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QUARTER 1: UNIT 1 Predicting Your Success: TED TALK, GRIT: The Power of Passion and Perseverance; submit a written response on to Google Classroom. Unit 1 & 2 Team Activity Form five to seven teams. Each team is responsible for one of the sectors of the external business environment discussed in the chapter (economic, political/legal, demographic, social, competitive, global, and technological). (Interpersonal, Resources, Information) <u>Quiz #1: Twenty core concepts from Unit 1 & 2</u> Critical Thinking Case Study-- UBER Hits a Bumpy Road.: Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your findings. -(Resources, Interpersonal, Information, Technology) Critical Thinking Case Study-- We Want Our MTV (International) <u>Quiz #2: Twenty Core Concepts on Unit 3 & 4</u> ...Continued Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your Findings -(Resources, Interpersonal, Information, Technology) Preparing for Tomorrow's Workplace Skills: - Do you have what it takes to be a successful franchisee? Working the Net: - Select three franchises that interest you.	QUARTER 2: Preparing Working the Net: Business Plan - Visit Sample Business Plans at http://www.bplans.com to review examples of all types of business plans. - Select an idea for a company in a field that interests you and, using information from the site, prepare an outline for its business plan. <u>Quiz #3- Comprehension of Twenty Core Concepts from Unit 5 & 6</u> Preparing for Tomorrow's Workplace Skills: Self Reflection (1 paragraph per question submitted into Google Classroom) Unit 7 Ethics Activity-Ethics Report; Training IT Replacement Working the Net Using a search engine, look for the term "company organizational charts," and find at least three examples of organizational charts for corporations, not-for-profits, or government agencies. Unit 8 Key Assignments: <u>Quiz #4-Comprehension of Twenty Core Concepts from Unit 7 & 8</u> <u>Critical Thinking Case Study- Discrimination in the Workplace Continues--submit on Google Docs</u> Semester Final Assessments: Part 1: Ethics Self-Assessment Part 2: Comprehensive Final on Key Concepts from Unit 1-8 - Students will be in 8 teams of 4 assigned to 1 Unit each. - They will choose the key concepts from each unit to teach and add to the final. (A form of Reciprocal Teaching)		

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Research them	
QUARTER 3: Preparing for Tomorrow's Workplace Skills- INTERVIEW - Are you motivated more by intrinsic rewards (satisfaction, sense of accomplishment, etc.) or by extrinsic rewards (money, bonuses, etc.)? Explain. - Interview some friends and classmates to find out what motivates them. Discuss your differences in perspective. (Interpersonal, Information) (Submit to Google Classroom through the provided Graphic Organizer. Team Activity - Assemble a team of three to five students. Imagine that you are the management team for a start-up business with limited resources but a need for a highly motivated, skilled workforce. <u>Quiz #5: Comprehension of Twenty Core Concepts from Unit 9 & 10</u> Preparing for Tomorrow's Workplace Skills Use the Google search engine, http://www.google.com, to conduct a search for "supplier information," and visit the websites of several firms (for example, Walmart, Northrop Grumman, Verizon, etc.). Compare the requirements companies set for their suppliers. How do they differ? How are they similar? Preparing for Tomorrow's Workplace Skills Pick a small business in your community. Preparing for Tomorrow's Workplace Skills Pick a specific product that you use frequently, such as a cosmetic or toiletry item, snack food, article of clothing, book, computer program, or video game. Using the Net -Considering a career in marketing? <u>Quiz #6- Comprehension of Twenty Core Concepts from Unit 11 & 12-Think of a product that you use regularly.</u> Working the Net-Visit Industry Week's website at http://www.industryweek.com. Under Archives, do a search using the search term "supply-chain management."	QUARTER 4: Preparing for Tomorrow's Workplace Skills Team Activity :Should companies outsource IT? Some executives believe that IT is too important to outsource and that application service providers don't have a future. Yet spending for ASP subscriptions, MSPs, and other forms of IT outsourcing such as cloud computing continue to grow. Working the Net--Learn more about the CERT Coordination Center (CERT/CC), which serves as a center of internet security expertise. Unit 15 & Unit 16 Key Assignment: -Build Key Portfolio Stock Project: your Investment Firm has just received \$100,000 to invest in 5 stocks FINAL (Introduced at the Start of 3rd QTR): Your Career in Business PART 1: PERSONAL REFLECTION: - How can you enhance your interpersonal skills? - Why is learning to plan so important in school and in the real world? - What skills should you develop in school that can transfer easily to your professional life and make it a success? - What are some strategies that will help you find, keep, and advance in your dream job? - What key attributes do employers look for when interviewing job candidates? PART 2: Student Companies FINAL ASSESSMENT: WORK-BASED LEARNING INTERNSHIP W/ AMAZON & OTHER PARTNERED LOGISTICS CENTERS

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"How many quizzes are there in the entire course?"

Six. Four are on page 15 and two are on page 16, so either page alone gives the wrong count.

Pages 15 and 16 · a course syllabus, quizzes outlined

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Why Locate the Failure?

The survey shows many ways to build these systems. But when a system answers wrongly, **different papers give different explanations for why**, and those explanations imply different fixes.

“How much did the leading region's revenue grow between the 2022 and 2023 tables?”

The system answers **14%**. The correct answer is 23%. What went wrong?

The page was read badly

The parser garbled a figure in the table, or the page was rendered too small to read it.

The right page was missed

Only one of the two tables was retrieved, so the comparison was never possible.

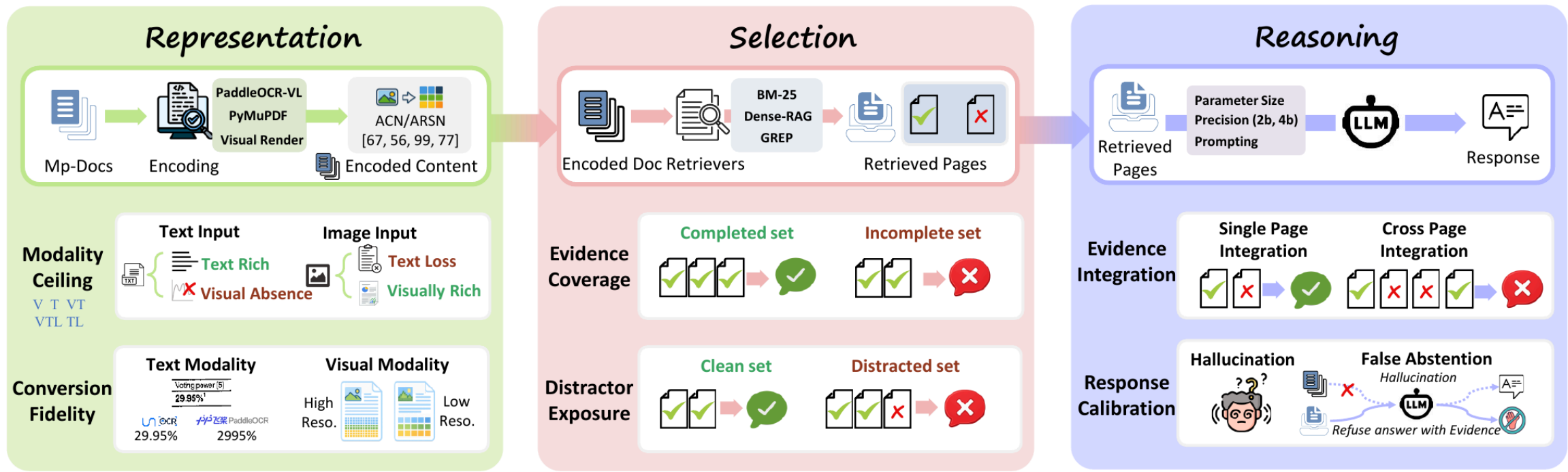
The model could not use it

Both tables were present and legible, and it still picked the wrong row or did the arithmetic wrong.

End-to-end accuracy cannot tell these apart, and **under a fixed compute budget you have to know which one to spend on**.

Three Failure Loci

Three places in the pipeline where an answer can go wrong, each with its own diagnostic question.



REPRESENTATION

Did we preserve the information the question needs from the page?

SELECTION

Did that information actually reach the model?

REASONING

Did the model use it correctly once it had it?

The order matters, and it is what makes attribution possible: **we cannot reason over evidence that was lost in encoding, or never selected.** So a locus can only be tested once the ones above it are neutralised.

How We Isolate a Failure

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

Encode

Every page becomes text, an image, or both.
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

Retrieve

▶ A retriever scores every encoded page against the question and returns the top k , restored to document order.

ColQwen3 · page-level

Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass. Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Every reported accuracy carries a confidence interval** from a document-level bootstrap, because questions from one document are not independent.

Representation Experiments

Two questions about the page itself: **what can an input express**, and **how much is lost converting the page into it**.

Modality ceiling

WHAT THE INPUT CAN EXPRESS

Four rungs of increasing richness. **T** is the PDF's embedded text layer. **TL** is layout-aware parser markdown. **TLV** adds the rendered page image. **V** is the page image alone.

The comparison asks whether the required evidence is still recoverable once the textual, structural, or visual signal is taken away, rather than assuming one representation is best everywhere.

Conversion fidelity

HOW MUCH SURVIVES THE CONVERSION

Hold the rung fixed and vary the quality of each channel instead: **the parser** for text, across PyMuPDF, MinerU, PaddleOCR-VL and Unlimited-OCR, and **the render resolution** for the image.

This separates what a modality cannot carry in principle from what a bad conversion destroyed in practice.

The annotated evidence pages are supplied in both, so **retrieval is not a variable here**. Whatever changes is the encoding.

Selection Experiments

Selection can fail in two opposite ways, and the point of the experiment is that **they are not equally damaging**. Both start from the pages known to contain the answer.

Evidence coverage

REQUIRED EVIDENCE REMOVED

Take questions that cite more than one gold page and **withhold one of them**. Pages are ranked within the document, so we can remove the highest-ranked gold page, the lowest-ranked, or keep only one.

A question that cannot satisfy a condition is excluded and recorded, never quietly satisfied with a different page set.

Distractor exposure

IRRELEVANT EVIDENCE ADDED

Keep the complete gold set and **add up to five top-ranked non-gold pages** alongside it. Conditions are blocked on the exact gold-page count, so questions needing different amounts of evidence are never pooled.

The supplied context stays at six pages or fewer, so distractor count is not confounded with context length.

The encoding and the reasoner stay fixed across both, so **the two effects can be compared directly** against the same oracle baseline.

Reasoning Experiments

With representation and selection controlled, what remains is the reasoner. Two different problems, so **they are tested separately**.

Evidence integration

CAN IT COMBINE WHAT IT HAS?

Split the questions by the **structure of the evidence** they need: one page and one source, several sources on one page, or evidence distributed across pages.

Then add a step-by-step reasoning prompt and see which of those groups it actually helps. The annotated pages are supplied throughout, so any deficit arises after the evidence has arrived.

Response calibration

SHOULD IT ANSWER AT ALL?

Vary both whether the evidence supports an answer and whether the model is **explicitly permitted to abstain**.

Answerable questions get their oracle gold pages. Unanswerable ones get the three top-ranked pages by BM25: plausible context that does not support an answer. That measures unsupported answers in one direction and false refusals in the other.

Everything is decoded greedily, so a condition is **reproducible from its identity** rather than from a seed.

Datasets Make Attribution Possible

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

MMLongBench-Doc

PRIMARY DIAGNOSTIC

135	47.5	847	244
documents	mean pages	answerable	unanswerable

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

LongDocURL

SECONDARY REPLICATION

396	85.6	2,325	8
documents	mean pages	questions	unanswerable

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

Text and Vision Complement Each Other

With the correct pages supplied, accuracy climbs the representation ladder: **29.0% text only, 42.6% image only, 49.5% with both.**

Evidence source	Accuracy				Abstention				Incorrect			
	T	TL	TLV	V	T	TL	TLV	V	T	TL	TLV	V
Text	36.9	44.4	51.0	42.3	23.0	11.5	7.3	8.0	40.1	44.1	41.6	49.7
Layout	26.3	33.9	49.2	43.2	20.3	10.2	0.8	4.2	53.4	55.9	50.0	52.5
Table	35.6	44.7	44.7	34.1	15.9	7.2	6.7	9.1	48.6	48.1	48.6	56.7
Chart	16.3	16.9	42.7	40.4	32.6	28.1	7.3	6.2	51.1	55.1	50.0	53.4
Figure	14.5	22.1	42.4	39.0	36.6	24.8	7.6	7.9	49.0	53.1	50.0	53.1
All	29.0	35.7	49.5	42.6	26.3	17.2	6.2	7.1	44.7	47.2	44.3	50.3

Answer outcomes by evidence source and representation rung, MMLongBench-Doc, oracle pages

– Text alone misses visual questions

At **T**, chart questions reach 16.3% and figure questions 14.5%. Adding the page image takes them to 42.7% and 42.4%.

– It fails by refusing, not by guessing

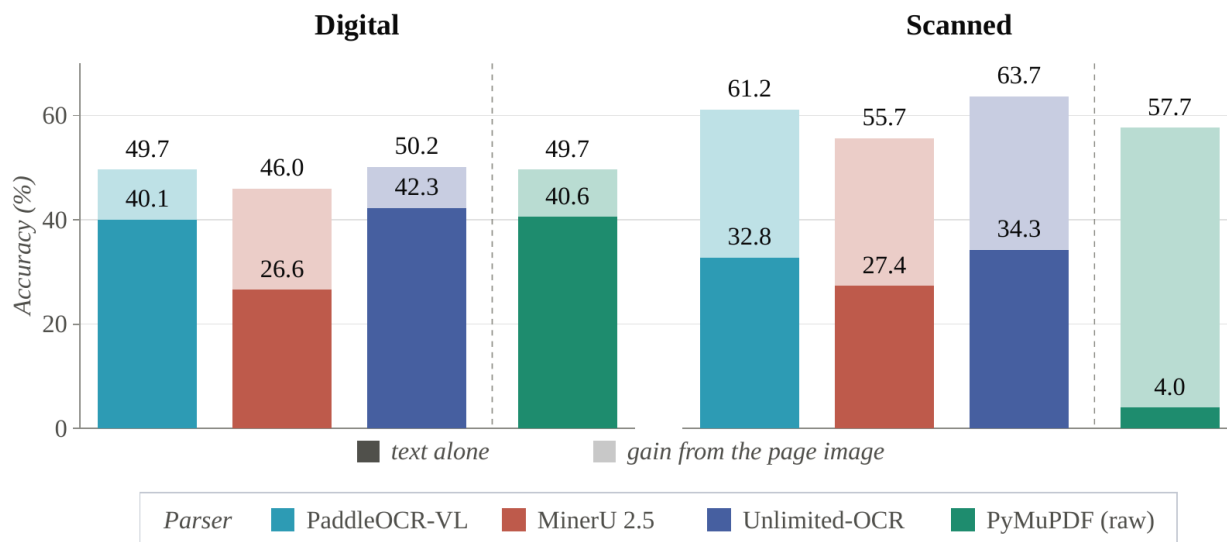
Abstention on figure questions falls from 36.6% to 7.6% across the ladder, while the incorrect rate stays broadly flat. The gain comes out of refusals.

– Images alone are weaker than both

V reaches 42.6% against 49.5% for **TLV**. Text exposes fine detail in a form that is easier to use than recovering it from pixels.

Conversion Quality Matters

A useful input type still fails if the page is converted badly. **Parser choice alone moves accuracy by tens of points**, so it is not neutral preprocessing.



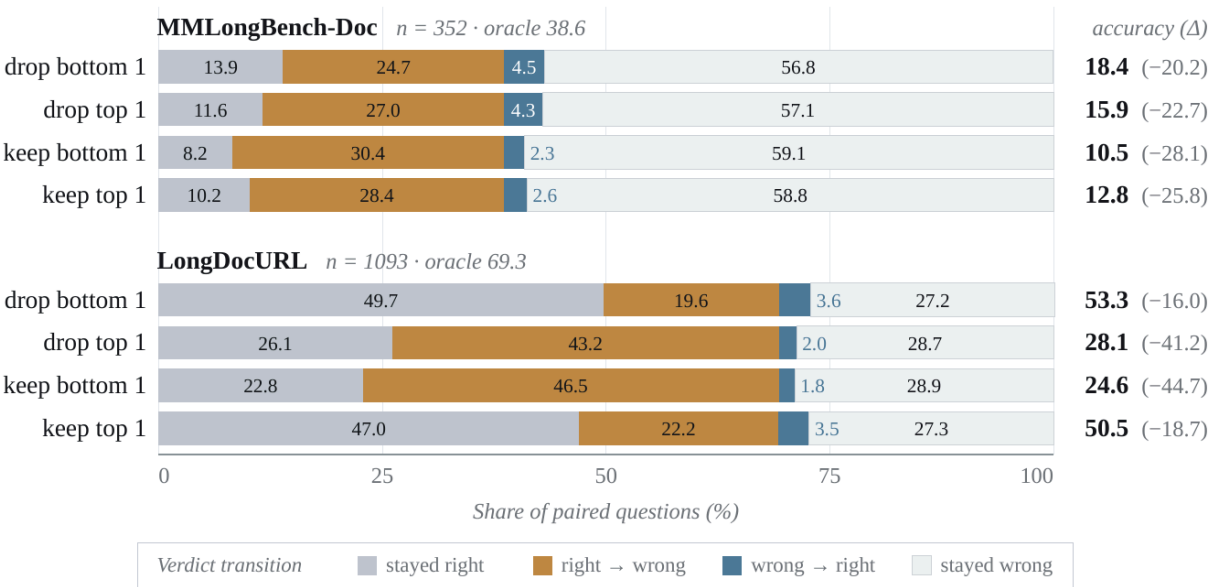
Parser accuracy by text source, with and without the page image

READING IT

- **Scans break the text channel entirely.** On scanned documents the PDF's embedded text layer is often absent, and accuracy collapses. Conversion failure does not merely degrade the representation, it removes it.
- **Parsers differ a lot.** MinerU is the weakest on both digital and scanned pages; PaddleOCR-VL and Unlimited-OCR recover substantially more.
- **Vision covers for the parser.** Adding the page image compresses the spread between parsers sharply, and the weakest parser benefits most. Many conversion errors are recoverable once a second channel still holds the signal.
- **Resolution is the mirror image.** Raising render resolution helps **V** far more than **TLV**, where the text already carries part of what the extra pixels would recover.

Missing Evidence Is Costly

Now remove one page the question needs. **Answers that were correct become wrong, and the reverse almost never happens.**



Verdict transitions after withholding or retaining gold evidence pages

READING IT

- Withholding one gold page breaks **24.7–27.0%** of previously correct answers, while only 4.3–4.5% are repaired. Keeping just one gold page widens it further: 28.4–30.4% break against 2.3–2.6% repaired.
- The loss is directional. This is more than the expected correlation between retrieval recall and accuracy. Removing known evidence produces a one-way loss that downstream reasoning rarely reverses.
- Which page survives matters too. Keeping the highest-ranked gold page preserves much more than keeping the lowest. Extractive questions tolerate partial evidence; counting and calculation do not.

Moderate Distraction Is Less Damaging

The opposite error: retrieval returns the right pages **plus a few extra**. Within the range we test, this costs far less than missing a required page.

Condition	MMLongBench-Doc						LongDocURL					
	T	TL	TLV	V	R→W	W→R	T	TL	TLV	V	R→W	W→R
<i>1 gold page</i>												
Oracle	37.1	44.6	63.7	55.0	—	—	65.6	74.0	74.7	64.5	—	—
+1 distractor	39.5	48.9	63.7	55.7	7.8	7.0	65.4	72.4	74.4	63.0	8.5	8.1
+2 distractors	38.8	47.9	64.3	56.1	8.2	8.0	65.1	68.7	72.0	63.9	9.3	7.5
+3 distractors	40.9	47.0	63.1	55.1	9.7	8.2	64.5	68.0	68.4	62.1	11.7	6.9
+4 distractors	41.9	45.3	62.5	52.5	9.5	6.6	63.5	68.0	72.2	62.5	10.5	8.7
+5 distractors	39.5	42.6	62.3	54.4	10.0	6.9	64.6	69.1	70.2	60.8	11.4	7.3
<i>2 gold pages</i>												
Oracle	28.9	33.3	37.8	34.6	—	—	62.2	63.4	70.4	61.5	—	—
+1 distractor	25.7	34.0	38.6	34.4	7.8	7.5	58.7	63.0	66.2	57.9	9.6	4.1
+2 distractors	25.3	33.6	35.7	32.4	9.6	6.0	58.9	62.2	66.4	57.4	10.9	4.4
+3 distractors	21.6	28.2	33.6	33.2	11.4	6.0	58.1	63.3	70.8	57.8	9.7	6.0
+4 distractors	24.6	30.7	43.7	31.3	11.8	11.8	57.8	63.2	68.3	57.1	13.3	6.1

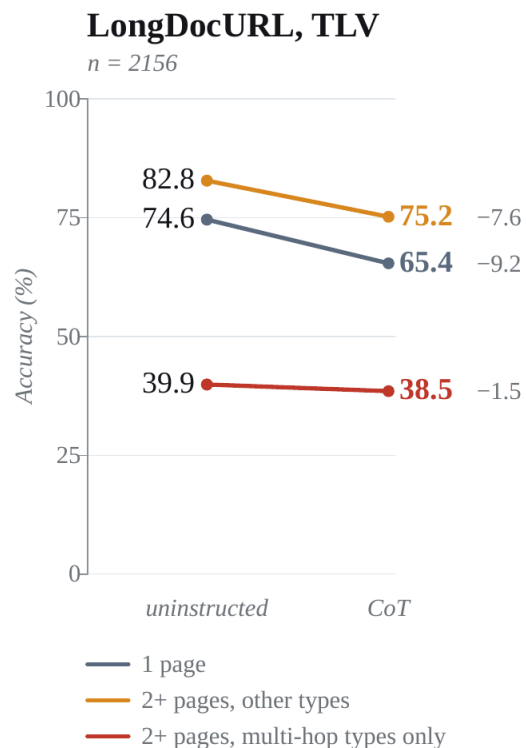
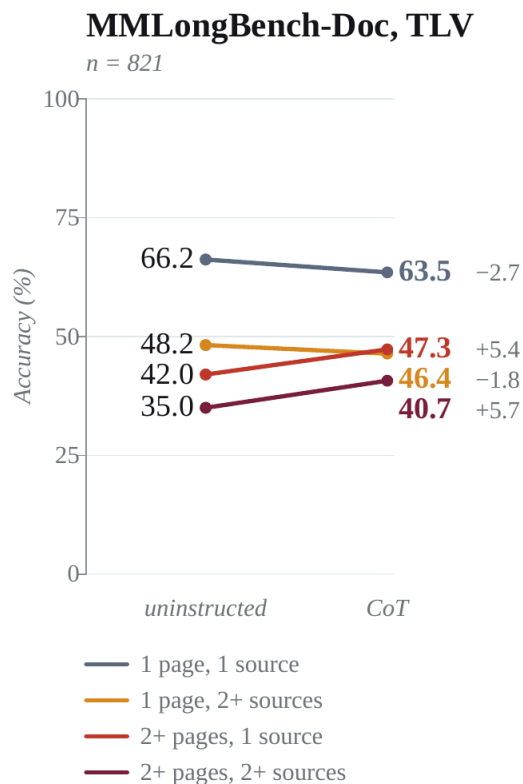
A complete gold set, then the same set plus one to five distractor pages. Tested range only: context stays at six pages or fewer

READING IT

- **Aggregate accuracy barely moves.** Single-page questions stay broadly stable, and multi-page questions decline gradually. LongDocURL degrades even less under comparable distractor counts.
- **But stability is not harmlessness.** The verdict transitions show added pages both breaking and repairing answers. Positive and negative effects partly cancel at the aggregate level.
- **A non-gold page is not always irrelevant.** Sometimes it adds a partial instance that makes the model overcount; sometimes it supplies the context that disambiguates the gold pages and repairs the answer.
- **So lean toward coverage** when the later context can absorb it, and control how the retrieved evidence is presented rather than minimising how much of it there is.

Reasoning Across Pages Remains Hard

Every correct page is supplied, in the richest representation. **Cross-page questions are still much harder**, and asking for step-by-step reasoning helps only some of them.



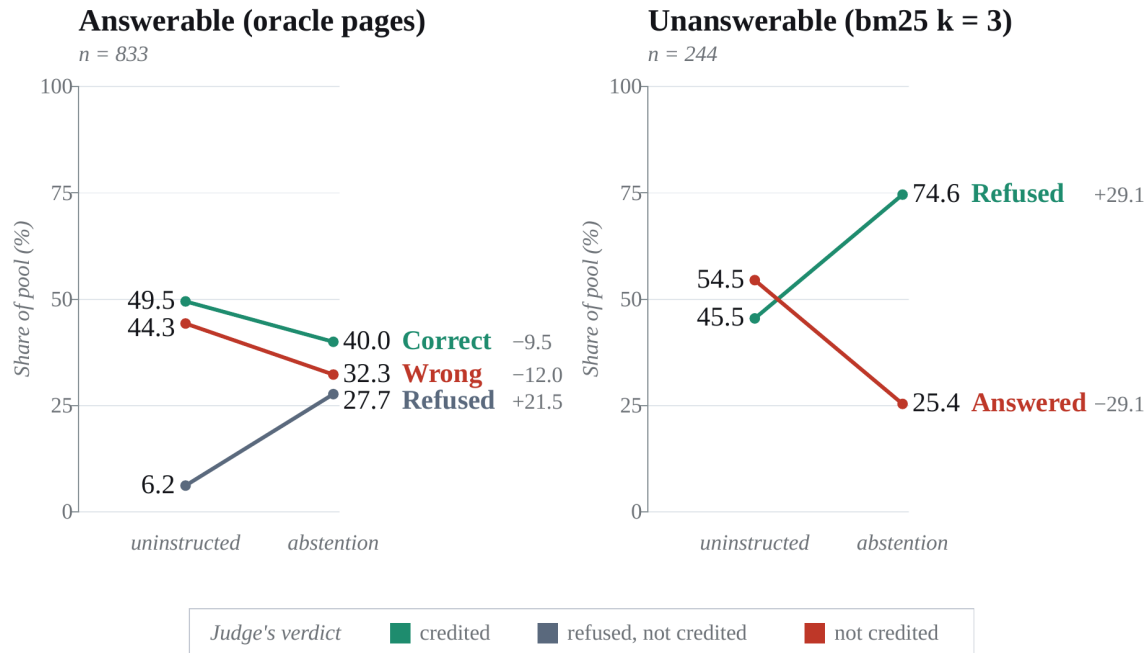
READING IT

- **The deficit is real.** On MMLongBench-Doc, one-source questions fall from 66.2% on a single page to 42.0% once the evidence spans pages, and to 35.0% when several sources must be combined as well.
- **Chain-of-thought helps precisely there.** It slightly hurts single-page questions (66.2 to 63.5), and lifts cross-page ones (42.0 to **47.3**, and 35.0 to **40.7**). The pooled integration deficit narrows from 24.6 to 18.2 points.
- **Most of the gain is structured evidence.** Tables improve by **10.9** points; charts, text and layout by two to three; figures fall slightly.
- **And it does not generalise.** On LongDocURL the same prompt reduces accuracy almost everywhere. More reasoning helps when it matches the task, not because there is more of it.

Effect of chain-of-thought across evidence structures, TLV rung, oracle evidence pages

Abstention Is Not Well Calibrated

The last reasoning decision is whether the evidence supports an answer at all. An abstention instruction makes the model more cautious, **but not more discerning**.



Effect of targeted abstention prompting at the TLV rung

READING IT

- **It works where it should.** On unanswerable questions, refusal rises from 45.5% to **74.6%**, cutting unsupported answers substantially.
- **And it fires where it should not.** On answerable questions, refusal rises from 6.2% to **27.7%**, and correct answers fall from 49.5% to 40.0%.
- **Difficulty is being read as insufficiency.** Under the instruction, refusal reaches 35.5% for figure questions and 30.3% for tables, against 24.1% for text, with oracle pages supplied, so the evidence is definitely there.
- **The prompt moves willingness, not judgement.** It shifts the operating point rather than separating sufficient evidence from insufficient.

A Simple Adaptive Method

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

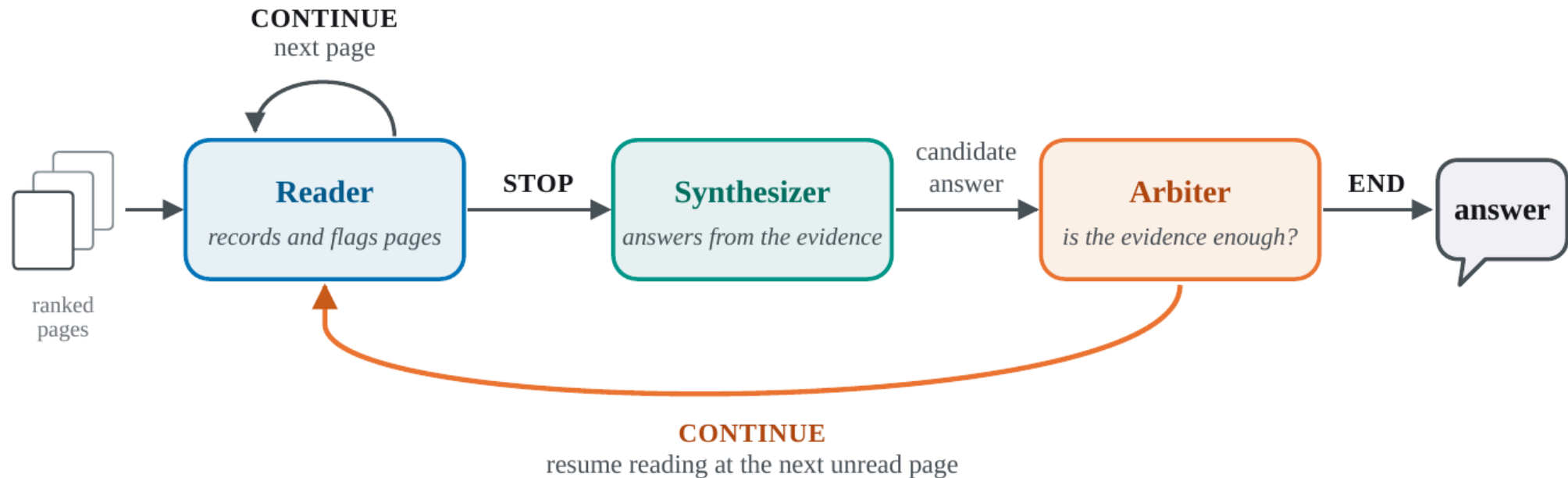
Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Reader, Synthesizer, Arbiter

One underlying model in **three roles**, differentiated only by the instruction and the input each receives.



reader

Works through the pages in retrieval-ranked order, recording answer-relevant evidence as it goes, and proposes stopping when it thinks it has enough.

synthesizer

Receives the **most important pages in full**, plus the short records from everything else read, and proposes an answer from that.

arbiter

Accepts the answer, or sends the reader back for more. It cannot write an answer itself, which makes **stopping too early recoverable** rather than final.

The loop is what makes the amount of reading **adapt to the question**: an easy one stops early, a hard one keeps going.

System-Level Validation, and Its Cost

On MMLongBench-Doc the method reaches **54.5%** under the official scorer, above the gold-page baseline using the same model family.

Method	OCR	Trained	Evidence source					Evidence pages			ACC
			TXT	LAY	CHA	TAB	FIG	SIN	MUL	UNA	
Official protocol: answer extraction + rule-based scorer											
Gold pages + VLM (TLV)	✓	×	48.1	49.6	39.4	50.5	42.5	57.7	37.4	52.5	49.5
<i>Qwen2.5-VL-7B</i>											
MDocAgent [†]	✓	×	43.1	31.2	32.6	38.7	37.9	53.5	23.8	28.3	38.5
MoLoRAG+	×	✓ ^a	42.7	38.8	33.3	38.5	40.7	52.9	27.6	35.9	41.0
MM-Doc-R1 [†]	✓	✓	46.2	38.1	40.8	52.8	35.9	56.0	31.2	68.2	49.7
<i>Qwen3-VL-8B</i>											
Whole document	×	×	—	—	—	—	—	46.9	35.3	25.8	38.5
MARDoc	✓	×	—	—	—	—	—	—	—	—	52.7
DocTrace	✓	✓	—	—	—	—	—	53.2	41.1	70.5	52.9
Ours (4-bit)	✓	×	48.0	41.9	43.7	59.9	41.8	58.9	37.8	71.9	54.5

MMLongBench-Doc accuracy, official protocol. Published systems on open 7-8B backbones, values taken from the cited papers

READING IT

- **Above the oracle baseline.** **54.5%** against **49.5%** for the same model reading only the annotated gold pages, which suggests useful context comes from **beyond** the pages the benchmark labels as evidence.
- **And above the published comparisons** on the same backbone, the strongest being 52.9%.
- **The gain is not uniform.** Multi-page questions reach 37.8%, still below DocTrace's 41.1%. Cross-page integration remains the hard part, exactly as the attribution study predicted.

8.2 GiB

mean peak VRAM

13.9

agent turns per question

9.3 min

median latency per question

16.6 min

mean, almost all decoding

Conclusion

Long-document understanding is, above all, a matter of **managing evidence well**.

Preserve

What survives the conversion of a page into text, pixels, and structure.

▶ Select

Which of it actually reaches the model, out of hundreds of pages.

▶ Reason

Whether the model combines it, and knows when it is not enough.

1 The survey

maps how systems preserve, find, and use information across pages.

2 The empirical study

shows where the failures arise, and that each one needs a different fix.

3 The simple method

shows those lessons can be turned back into a working system.

Thank you, and happy to take questions.